lúbadh

_travelers guide
_for navigating time loops

2	10	4
decks · stereo link	min per loop, per deck	taps per deck

[rec] press [rec] hold **[shift]** = retrig/shift opp. = the other deck's button › then
◆ = preset / mode only

every control is per deck: left and right are identical

combos shown for one deck · mirror them for the other · ◆ your own preset: sheet 2 ↓

amber = hold

01 transport — lbh_01

[rec]	record › press again to stop & play. press while playing = **overdub**
[erase]	erase this deck's loop
[shift]	jump back to start
turn **[speed]**	speed & direction: center = stall, cw = forward to 4×, ccw = reverse to 4×
[shift] + turn **[speed]**	knob is muted until release, then jumps straight to the new speed
[link] up	right deck mirrors left · saves and loads stereo
[shift]	adds a playhead at start (up to 4)
◆ multi-tap presets · off in tape-looper	

02 recording — lbh_02

[erase] + **[rec]**	**punch-in**: replaces audio while recording
[shift] **[shift]** + **[rec]**	**input monitoring**: on (rec lit) › armed (rec pulses) › off
armed **[rec]**	arm › record › stop & play · **[erase]** disarms, loop kept
[shift] + opp. **[rec]**	looping ← → **one-shot record** (records one pass, then stops)
[rec] jack	trig starts / stops recording, like **[rec]** (latching)
[rec] jack	gated: records only while the gate is high
◆ rec jack = gated · off in tape-looper	

* overdub keeps the old layer at 0.9 by default, so layers fade a little each pass (dub level, 04).

__fig.01 : speed

white speed LEDs = notches



4× rev	½ rev	stall	½	1×	2×	4×
+24 st	−12 st	—	−12 st	0 st	+12 st	+24 st

* speed sets pitch: every doubling = +1 octave · reverse plays backwards

03 playback — lbh_03

[shift] + opp. **[erase]**	looping ← → **one-shot playback** (rec blinks white / mixed)
[shift] or **[retrig]** jack	one-shot: play one pass of start › length
touch the reel	warble or stall the tape · trimmer sets sensitivity
[speed] cv	±5 v, summed with the knob
[erase] + opp. **[erase]**	flip playback direction · speed LEDs blink
◆ v/oct speed mode only · off in tape-looper	
[speed] cv	tracks 1v/oct: sequence the loop's pitch
◆ v/oct · seq presets · off in tape-looper	

04 time knob — lbh_04

[shift] + **[rec]**	cycle the time function (amber LED by **[clk out]** shows which)
LED off	**clock divisions**: triggers per loop at **[clk out]**: 1–12, 16, 24, 32, 64
LED solid	**dub level**: how much old audio survives each overdub. high = delay feedback
LED pulsing	**user parameter**: crossfade, tape or slew (preset picks)



05 start · length — lbh_05

turn **[start]**	where the loop begins inside the recording
turn **[length]**	how much plays before jumping back. tiny = wavetable-like tones
[shift] + turn **[time]**	**quantise** start & length: lit LED sections = 1/2, 1/4, 1/8 …
[shift] + turn **[start]**	fine move that ignores the quantise grid (same for **[length]**)



* locks = quantise grid (1/8)

06 save · load — lbh_06

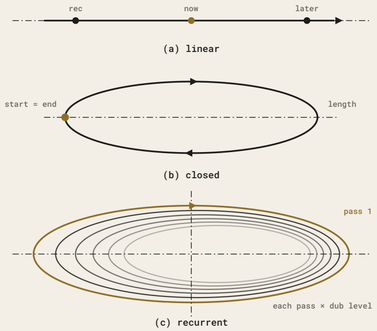
[shift] **[erase]** + **[rec]**	**save** to usb › saved (folder 0, newest first)
[shift] + **[erase]**	**load menu**
in the menu	own **[time]** = loop · opposite **[time]** = folder · **[rec]** loads · hold **[shift]** previews 5 s · **[erase]** exits
[link] up	saves and loads stereo (mono loads of stereo are summed)
power-on	last session reloads automatically

07 presets — lbh_07 · v2.1

[shift] **[shift]** + **[erase]**	**preset menu** for that deck
turn **[time]**	pick 1 of 12 (shown on the LED ring) · **[rec]** loads · **[erase]** cancels
usb › presets	.txt files, sorted 01–12. build your own with the preset editor

__fig.02 : the return

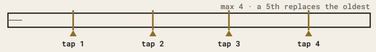
mm_02



08 multi-tap — lbh_08

◆ multi-tap presets only

[shift]	new playhead from start
[erase] + **[shift]**	delete the taps
[erase] trig, cv exp.	removes taps remotely



09 sync · clock — lbh_09

[clk out]	triggers per loop (set by clock div)
[retrig] jack	resets the playhead to start: sync to an external clock
[rec] jack	external clock sets the loop length
◆ clockable preset · off in tape-looper	
[erase] + [rec]	tap tempo instead of punch-in
◆ erase + rec = tap tempo · off in tape-looper	

10 aux · routing — lbh_10

aux in fader	sends **[aux]** in to the left ← → right deck input (+ **[cv]** ±5 v)
aux out fader	blends left ← → right outputs at **[aux]** out (+ **[cv]** ±5 v)
rear switch	aux out fader pre or post the output level, per deck



11 reset · calibrate — lbh_11

[shift] **[shift]** + **[rec]** **[rec]**	**reset modes** to default · audio kept · speed LEDs pulse
power on, hold **[shift]**×2 **[erase]**×2	**calibration**: speed knobs min, 0 v › **[rec]** each deck, then 1 v and 3 v; knobs max, no cv › **[rec]** = 5 v
amber blinks	voltage out of range: defaults used

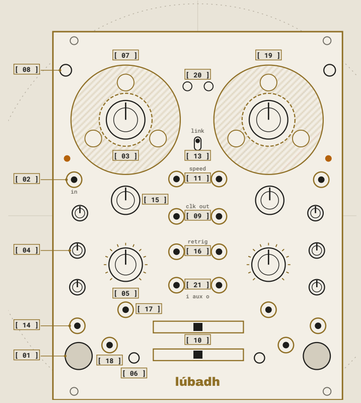
__fig.03 : front panel

01–11 = sections · 13–21 = hardware

left deck labelled · right deck mirrors it

[01]	**__rec :** record · overdub · stop
[02]	**__in :** input + level
[03]	**__speed :** rate & direction
[04]	**__time :** clock div · dub · user
[05]	**__start :** loop start
[06]	**__erase :** erase · save / load
[07]	**__LED ring :** position · presets
[08]	**__retrig/shift :** restart · shift
[09]	**__clk out :** clock triggers
[10]	**__aux faders :** in / out crossfade
[11]	**__speed cv :** ±5 v · 1v/oct

[13]	**__link :** stereo mirror
[14]	**__out :** output + level
[15]	**__length :** loop length
[16]	**__retrig :** jack, restart
[17]	**__cv :** aux fader cv
[18]	**__rec :** jack, record gate
[19]	**__reel :** capacitive touch
[20]	**__speed LEDs :** notches
[21]	**__aux :** in · out jacks



_where things hide

12 · lbh_12

preset files

usb › presets › .txt · instruo preset editor

- **speed** record variable · fixed, play follow · hold
- **speed control** notched · stepped · smooth · v/oct, markers 0–4
- **multi-tap** engine off · on
- **time user** crossfade · speed slew · tape emulation
- **rec jack** latching · gated · external clock
- **ext clock** average, resolution, timeout, divisions
- **quantise** all · even · odd · powers of two
- **erase + rec** punch-in · tap tempo
- **timing** min loop, crossfade, retrig delay, max dub
- **tape** wow / flutter, crinkle, age, hysteresis, wear, saturation, reverb
- **input filter** low cut freq + q, high cut
- **touch** off · stall · dip

usb disk

- **samples** 12 folders × 12 loops
- **saved** folder 0, newest first
- **presets** first 12 show in the menu

tails & tags

250 ms, in the file name

- {s} silent tail (default): keeps timing
- {n} no tail: smoothest, timing loosens
- {c} copy tail: keeps timing, may click

hardware

- **cv expander** **[length]** **[start]** **[time]** cv + **[erase]** trig, per deck
- **rear pcb** touch on / off · pre / post output level
- **front** touch sensitivity trimmer

factory presets

ring = what the LED ring shows in the preset menu

- **01 tape looper** · tape feel · varispeed rec
- **02 clean** · crossfade on time · little tape
- **03 classic** · v1 behaviour · fixed rec speed
- **04 multi-tape** · multi-tap delay lines
- **05 clockable** · sync to an external clock
- **06 octave delay** · pitch-shifting delay line
- **07 seq mono** · 1v/oct · retrig delay
- **08 seq poly** · multi-tap 1v/oct · speed hold
- **09 melontron** · 4-note mellotron chords
- **10 broken tape** · maxed-out warble & grit

your preset

_travelers guide
_for decoding your own time loops

0	off	notched
changed from tape-looper	multi-tap	speed control

drop a preset .txt › read in your browser, never uploaded share link download .txt

◆ = changed from tape-looper

active preset: **Tape-Looper** · factory · 01-Tape-Looper

settings from the lúbadh v2.1 preset editor · instruo

ghosted = not used by this preset

01 load a preset

lpr_01

install it	copy the .txt into the **presets** folder on the usb · sorted a–z, the first 12 appear in the menu
[shift][shift] + [erase]	preset menu for that deck · [time] picks 1–12 · [rec] loads · [erase] cancels

02 factory presets

lpr_02 · manual p.28

01 tape-looper	close to the original lúbadh, with vari-speed recording and the time pot on tape amount
02 clean	vari-speed recording, the time pot on crossfade, most tape emulation off · values coming soon
03 classic	mimics the v1 firmware · vari-speed recording off · values coming soon
04 multi-tape	multi-tap delay lines · values coming soon
05 clockable	built for an external clock · chopping loops, beat repeats · values coming soon
06 octave-delay	a pitch-shifting octave delay line · values coming soon
07 sequencingmono	1v/oct pitch plus a retrigger delay so pitch lands before the note · values coming soon
08 sequencingpoly	multi-tap SequencingMono with playback speed on hold · polyphonic loop playback · values coming soon
09 melontron	a knock-off mellotron · 4-note polyphony from multi-taps and 1v/oct · values coming soon
10 broken-tape	maxed-out tape emulation for warbled distortion · values coming soon

03 speed

lpr_03

record speed	0 · variable recording follows the speed knob, like a vari-speed tape deck
play · looping	0 · follow playback follows the speed knob
play · one-shot	0 · follow playback follows the speed knob
speed control	0 · notched the speed LEDs light the markers · the knob ramps between them · −4× to 4×
speed markers	0× · 0.5× · 1× · 2× · 4× knob notches, forward and reverse · 0 is always added
speed slew	25 ms tape-like lag and pitch whips on speed changes

__fig.04 : the speed knob

this preset



notched: ramps between markers · −4× to 4×

08 touch plate

lpr_08

touch plate	0 · off the touch plate does nothing

04 time · taps

lpr_04

time pot · user	2 · tape emulation amount the user time mode scales the tape emulation (wow, flutter, crinkle …)
crossfade	250 ms crossfade used whenever the time pot isn't setting it
multi-tap	0 · off · single-tap a new tap fades the old one out: one seamless playhead
max dub level	0.9 the top of the dub level · old layers fade a little each pass
min loop length	1280 samp ≈ 26.7 ms loop length with the length pot at minimum · short = karplus / granular, long = smoother
retrig delay	0 ms off · retrig fires immediately

05 rec jack · clock

lpr_05

rec jack	0 · latching record a trigger toggles record, like the button
clock average	4 pulses averaged · higher = steadier, slower to follow tempo changes · only with rec jack = external clock
clock resolution	64 clock pulses per full loop · higher = finer sync · only with rec jack = external clock
clock timeout	0 s off · waits for the next pulse however long it takes · only with rec jack = external clock
clock divisions	0 · all [1–12, 16, 24, 32, 64] the clock-out and sync divisions the time knob steps through
start · length quant	3 · powers of two [2, 4 … 64] start and length snap to these divisions of the loop · hold shift for an unquantised nudge
erase + rec	0 · punch-in record erase + rec overwrites part of the loop

06 tape

lpr_06 · white tick = tape-looper

wow · flutter	0.5 slow (wow) and fast (flutter) speed wobble
crinkle	0.2 irregular jolts, like a creased tape loop
tape age	0.3 bandwidth loss · 0 = fresh and crisp, 1 = muffled
hysteresis	0.2 magnetic smudging of the input
wear	0.125 quiet parts fade over passes, loud parts survive
reverb	0.6 a one-knob reverb
saturation knee	0 tanh-like soft saturation
compensation	0.2 drives the tape hotter, a little bass lift · above 0 can distort as dubs build

07 input filter

lpr_07 · fig.05

low cut	20 Hz resonant 2-pole low cut · the tape-machine bass bump
low cut q	0.2 gentle
high cut	15000 Hz 1-pole high cut · lower = darker, older tape

__fig.05 : input filter band

documented values only



low cut 20 Hz · 2-pole, resonant · −12 dB/oct below
high cut 15000 Hz · 1-pole · −6 dB/oct above
q 0.2 · instruo: 0 off, 0.4 – 0.6 resonant, over 0.6 too much · peak not dr

_in plain words

what this preset is for

- ◆ a straight looper: one playhead, the speed knob ramping between notches
- ◆ the user time mode controls tape emulation amount
- ◆ some tape colour

_the file

as it would download · red = changed from tape-looper

RecordSpeed	0
Playback.Looping	0
Playback.Oneshot	0
MultiTap	0
SpeedControl	0
SpeedMarkers	0, 0.5, 1, 2, 4
TimePot	2
RecordJack	0
Clock.Average	4
Clock.Resolution	64
Clock.Timeout	0
ClockDivisions	0
Quantisation	3
EraseRecord	0
MinLength	1280
CrossfadeDuration	250
RetrigDelay	0
MaxDubLevel	0.9
WowFlutterDepth	0.5
CrinkleDepth	0.2
TapeAge	0.3
Hysterisis	0.2
Wear	0.125
Reverb	0.6
Knee	0
Compensation	0.2
LowCutFreq	20
LowCutQ	0.2
HighCutFreq	15000
SpeedSlewTime	25
CapacativeTouchMode	0

_how it works

lpr_09

- ◆ read in your browser, never uploaded · this browser remembers the last preset
- ◆ sheet 1 follows along: its ◆ preset-only rows say whether they're on
- ◆ share link packs every setting into the address · download writes a lúbadh .txt